# AP-E4 Same-Soliton Audit: Conditional $N = 2$ Moduli-Space SQM, Families-Callias Determinant Line, CPT/Anti-Canonical Polarization, and Gauge-Basic WZW Transgression

Route F research note

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**Abstract**

This note asks the stronger AP-E4 question that the independent vortex model cannot answer: can a physical tangent fermion, its Callias determinant line, and the level-two WZW line all be derived on the *same* charged two-colour $B = 1$ soliton? The answer is a rigorous conditional theorem and a rigorous underdetermination result, not a promotion claim.

At two-derivative adiabatic order, a one-multiplet tangent $N = 2$ SQM is characterized at low energy by a smooth normalizable bosonic zero-mode family, a uniform gap, a Fredholm fermion kernel $\mathcal{K}_F \simeq T^{1,0}\mathbb{CP}^1 = \mathcal{O}(2)$, the $N = 2$ Ward identities, and the Levi–Civita connection. Exact microscopic supercharges are a sufficient mechanism for these identities, but not a logical necessity because accidental/emergent worldline supersymmetry is possible. Our conservative same-model audit therefore accepts either microscopic supercharges or an independent derivation of all emergent Ward identities. The present charged two-colour proxy specifies a bosonic potential and selected Hessian blocks, but no supersymmetry transformations and no Yukawa mass family. Therefore a gapped trivial fermion family and a topologically winding one are both compatible with the current bosonic evidence. The same-model tangent fermion and Callias index are underdetermined.

For a declared rank-one asymptotic Callias eigenbundle over $S^2_\infty \times \mathbb{CP}^1$ with $c_1(\mathcal{E}_+) = px + qy$, the families boundary push-forward gives, in the stated chirality convention,

$$\operatorname{rank}\operatorname{Ind}\mathscr{D} = \epsilon p, \qquad c_1(\det\operatorname{Ind}\mathscr{D}) = \epsilon pq\, y.$$

Thus $\epsilon p = 1, q = 2$ is the minimal rank-one template for $\mathcal{O}(+2)$. A gauge-invariant discrete Berry computation gives $c_1 = 1.999999999999981$, while a finite $3 \times 3$ template/control kernel-family Hamiltonian has residual below $10^{-15}$ and unit nonzero gap. These are regressions of the conditional topology, not substitutes for the absent four-dimensional Callias operator.

In a fixed Dolbeault polarization, CPT must use Serre duality:

$$E_- = K_{\mathbb{CP}^1} \otimes E_+^\vee = \mathcal{O}(-4), \qquad (h^0, h^1) = (0, 3).$$

The naive raw inverse $\mathcal{O}(-2)$ has only one negative-chirality zero mode. The necessary anti-canonical regulator factor is not derived by the current mother model.

Finally, the degree-five mixed-WZW character is correctly transgressed along the spatial three-cycle,

$$\widehat{\mathcal{L}}_B = \int_{\Sigma_3} \mathrm{ev}^* \,\widehat{\Omega}_5 \in \widehat{H}^2(\widetilde{\mathcal{C}}_B; \mathbb{Z}).$$

After an equivariant differential refinement and globally trivial vertical holonomy establish gauge-basic descent, its pullback to the same soliton moduli sphere is $\mathcal{O}(+2)$ exactly when $\int_{\Sigma_3 \times \mathbb{CP}^1} \mathrm{ev}^* c(\widehat{\Omega}_5) = 2$. The factorized conditional map gives $nB \deg(s) = 2$, but the present proxy has not identified its scalar $S^2$ with the soliton collective sphere and has not proved equivariant descent. The AP-E3/AP-E4 composition gate and the degree-one Route-E portal therefore remain closed.

# Contents



# 1 Scope, logic, and strongest defensible conclusion

There are three spheres that must not be identified by notation alone:

(S1) the $SU(2) \simeq S^3$ Skyrme field whose spatial degree is the baryon number $B$;

(S2) the charged-scalar vacuum direction $s \in S^2$ entering the mixed WZW target $S^3 \times S^2$;

(S3) a putative normalizable collective-coordinate sphere $\mathcal{M}_B \simeq \mathbb{CP}^1$ of the *same* $B = 1$ soliton.

An abstract diffeomorphism $S^2 \simeq \mathbb{CP}^1$ does not identify (S2) and (S3). Such an identification requires an explicit family of solutions and an evaluation map. The current nonlinear proxy establishes a product-target $B = 1$ saddle and selected positive radial blocks. It does not establish a normalizable $\mathbb{CP}^1$ degeneracy of that saddle, a four-dimensional supersymmetric completion, or any fermionic fluctuation operator.

We therefore distinguish three kinds of statement:

**Definition.** A construction that exists once its displayed geometric inputs are given, such as differential-cohomology fiber integration.

**Conditional theorem.** An implication with every sufficient audit hypothesis displayed explicitly; necessity is claimed only for the stated low-energy one-multiplet data, not for a microscopic supersymmetry mechanism.

**Same-model derivation.** A conclusion obtained from the actual charged two-colour microscopic action and its $B = 1$ solution.

Only the first two are achieved here.

**Theorem 1.1** (Strongest current AP-E4 conclusion)**.** *The present charged two-colour bosonic data neither imply nor exclude a physical tangent fermion. They admit a precise conditional completion in which the same-soliton Callias kernel is $\mathcal{O}(2)$, the $B = +1$ SQM has three positive-chirality ground states, the fixed-polarization $B = -1$ sector uses $\mathcal{O}(-4)$, and the spatially transgressed WZW line has degree $+2$. None of these conditional inputs is selected by the current fermion/Yukawa or gauge-equivariant data. Consequently*

$$\texttt{ap\_e3\_ap\_e4\_composition\_gate\_closed=false}, \qquad \texttt{degree\_one\_route\_e\_portal\_built=false}.$$

# 2 Same-soliton collective-coordinate expansion

## 2.1 Configuration space and horizontal zero modes

Let $\mathscr{S}[\Phi, \Psi]$ be a hypothetical four-dimensional mother action whose bosonic sector contains the charged two-colour candidate. A family of static unit-soliton solutions is a map

$$\Phi_1 : \Sigma_3 \times \mathcal{M}_1 \longrightarrow \mathscr{F}, \qquad m \longmapsto \Phi_1(\cdot; m), \qquad B[\Phi_1(m)] = 1, \tag{1}$$

where $\mathscr{F}$ is the field-value space. Gauge equivalence is removed by a horizontal condition. If $m^a$ is a local complex coordinate and $\epsilon_a$ is the compensating gauge parameter, define

$$\delta_a^H \Phi_1 := \partial_a \Phi_1 - \delta_{\epsilon_a} \Phi_1, \qquad \mathscr{G}_\Phi(\delta_a^H \Phi_1, \delta_\lambda \Phi_1) = 0 \quad \text{for every } \lambda. \tag{2}$$

The physical moduli metric is the Hessian kinetic pairing

$$G_{a\bar{b}}(m) = \int_{\Sigma_3} \mathrm{d}^3x\, \mathscr{G}_{\Phi_1(m)}\big(\delta_a^H \Phi_1, \delta_{\bar{b}}^H \Phi_1\big). \tag{3}$$

Three requirements are separate: $\delta_a^H \Phi_1$ must solve the linearized equations, be $L^2$-normalizable, and have positive finite norm. The current radial Hessian does not prove any of them for an orientational mode.

Assume for the moment that $\mathcal{M}_1 = \mathbb{CP}^1$. On charts $U_N = \{w\}$, $U_S = \{v\}$ with $v = w^{-1}$, symmetry and direct chart matching give

$$G_{w\bar{w}} = \frac{C}{(1 + |w|^2)^2}, \qquad C > 0, \qquad G_{v\bar{v}} |\mathrm{d}v|^2 = G_{w\bar{w}} |\mathrm{d}w|^2. \tag{4}$$

The number $C$ must be computed from Eq. (3) for the same charged soliton; borrowing it from a vortex does not close the gate.

## 2.2 Fermion zero-mode family

Let the quadratic fermionic action on $\Phi_1(m)$ be

$$S_F^{(2)} = \int \mathrm{d}t\, \mathrm{d}^3x\, \overline{\Psi}\big(\mathrm{i}\partial_t - \mathscr{D}_m\big)\Psi, \qquad \mathscr{D}_m = \Gamma^i \nabla_i + \mathscr{Y}(\Phi_1(x; m)). \tag{5}$$

Here the representation, gauge connection, grading, and Yukawa map $\mathscr{Y}$ are microscopic data. If the kernel dimension is constant and the rest of the spectrum has a uniform gap, the kernels form a smooth Hermitian bundle

$$\mathcal{K}_F := \bigsqcup_{m \in \mathcal{M}_1} \operatorname{Ker} \mathscr{D}_m \longrightarrow \mathcal{M}_1. \tag{6}$$

For an orthonormal local frame $\Xi_A(x;m)$, the adiabatic expansion $\Psi = \chi^A(t)\Xi_A(x;m(t)) + \Psi_\perp$ gives the Berry connection

$$(\mathcal{A}_a)^A{}_B = \int_{\Sigma_3} \mathrm{d}^3x\, \Xi_A^\dagger \partial_a^H \Xi_B, \qquad D_t\chi^A = \dot{\chi}^A + (\mathcal{A}_a)^A{}_B \dot{m}^a \chi^B. \tag{7}$$

**Theorem 2.1** (Low-energy characterization and sufficient microscopic closure)**.** *Fix the two-derivative adiabatic truncation, demand one complex bosonic coordinate, one complex Grassmann partner, no extra zero modes, and no superpotential. The low-energy tangent SQM requires the geometric/kernel content (T1), (T3)–(T5) and the $N = 2$ Ward identities in (T2). A sufficient same-model microscopic closure is furnished when all of the following hold:*

*(T1) Eq.* (1) *gives a smooth normalizable* $\mathbb{CP}^1$ *family with positive metric and a uniform noncollective bosonic gap;*

*(T2) either the microscopic theory has two real supercharges preserved by the soliton and maps each $\delta_a^H\Phi_1$ to a normalizable fermion zero mode, or an independent calculation derives the complete emergent worldline $N = 2$ Ward identities;*

*(T3) Eq.* (5) *is a uniformly Fredholm family and $\mathcal{K}_F \simeq T^{1,0}\mathcal{M}_1$ as a Hermitian holomorphic line bundle;*

*(T4) the induced Berry connection under this isomorphism is the Levi–Civita connection, and the Ward identities fix the curvature four-fermion term;*

*(T5) all modes orthogonal to the collective multiplet remain uniformly gapped.*

*Proof.* At low energy, a physical collective coordinate requires (T1); a worldline superpartner and its closure require the Ward-identity content of (T2); tangent-valued transition functions require (T3); covariance of the supercharge requires (T4); and a local adiabatic truncation requires (T5). This does not make exact microscopic supercharges necessary: the same Ward identities may arise accidentally after integrating out gapped modes.

For the stated sufficient closure, choose complex coordinate $m$ and transport the normalized kernel frame through the isomorphism $\mathcal{K}_F \simeq T^{1,0}\mathcal{M}_1$. Conditions (T1)–(T5) give the standard covariant action

$$L_{N=2} = G_{m\bar{m}}\dot{m}\dot{\bar{m}} + \mathrm{i}G_{m\bar{m}}\bar{\chi}^{\bar{m}}D_t\chi^m - \frac{1}{2}R_{m\bar{m}m\bar{m}}\chi^m\bar{\chi}^{\bar{m}}\chi^m\bar{\chi}^{\bar{m}}, \tag{8}$$

$$D_t\chi^m = \dot{\chi}^m + \Gamma^m{}_{mm}\dot{m}\chi^m. \tag{9}$$

Up to conventional factors, the transformations

$$\delta m = \varepsilon\chi^m, \qquad \delta\chi^m = -\mathrm{i}\bar{\varepsilon}\dot{m} - \Gamma^m{}_{mm}(\delta m)\chi^m \tag{10}$$

and their conjugates close on time translation modulo the equations of motion. This is precisely the one-complex-dimensional tangent $N = 2$ sigma model [1, 2]. □

On $v = w^{-1}$, a tangent fermion must obey

$$\chi^v = \frac{\partial v}{\partial w}\chi^w = -w^{-2}\chi^w. \tag{11}$$

The transition has degree two in the line-bundle convention, hence $\mathcal{K}_F \simeq T^{1,0}\mathbb{CP}^1 = \mathcal{O}(2)$. The verifier checks the metric, fermion norm, and affine-connection transformation in both charts.

**Theorem 2.2** (Bosonic underdetermination)**.** *The bosonic soliton family, its classical Hessian, and the mixed-WZW coefficient do not determine $\mathcal{K}_F$ or even* $\operatorname{rank}\mathcal{K}_F$*.*

*Proof.* Hold the bosonic action fixed. Add a spectator vectorlike fermion with a constant invertible gauge-invariant mass $M$. Its spatial operator has no protected zero mode and its kernel bundle is zero. Alternatively choose an allowed Yukawa mass whose normalized asymptotic eigenspace winds over $S^2_\infty \times \mathcal{M}_1$; the Callias boundary class below may then have nonzero index and determinant Chern class. These fermionic choices leave the tree-level bosonic action and Hessian unchanged. Therefore no function of the bosonic data alone selects between them. □

The present charged two-colour theory is not supplied as a four-dimensional supersymmetric action. Emergent worldline supersymmetry is logically possible, but would require the same equalities of metrics, Berry connections, and interactions in (T1)–(T5); it cannot be inferred from an abstract moduli-space complex structure.

# 3 Families-Callias index and determinant line

## 3.1 Uniform Fredholm condition

A Callias operator has the schematic graded form

$$\mathscr{D}^+_m = D^+_{A(m)} + \mathscr{Y}_m : H^1(\Sigma_3, E^+) \longrightarrow L^2(\Sigma_3, E^-). \tag{12}$$

For every $m \in \mathcal{M}_1$, require constants $R, \mu > 0$, independent of $m$, such that outside the radius $R$

$$\mathscr{Y}^\dagger_m \mathscr{Y}_m \geq \mu^2, \qquad \big\|[D_{A(m)}, \mathscr{Y}_m]\big\| \leq \mu^2 - \delta \quad (\delta > 0). \tag{13}$$

These hypotheses make the family Fredholm and define

$$\mathrm{Ind}\, \mathscr{D}^+ = [\mathrm{Ker}\, \mathscr{D}^+] - [\mathrm{coker}\, \mathscr{D}^+] \in K^0(\mathcal{M}_1), \tag{14}$$

with determinant line

$$\lambda_{\mathrm{C}} := \det \mathrm{Ind}\, \mathscr{D}^+ = \det \mathrm{Ker}\, \mathscr{D}^+ \otimes (\det \mathrm{coker}\, \mathscr{D}^+)^\vee. \tag{15}$$

The Callias theorem and its families generalization reduce this class to the positive-mass eigenbundle at spatial infinity [3, 4, 5].

## 3.2 Boundary push-forward and the required class

Let $x$ and $y$ be positive generators of $H^2(S^2_\infty; \mathbb{Z})$ and $H^2(\mathbb{CP}^1; \mathbb{Z})$, respectively, normalized by unit integral. Consider the minimal rank-one product template

$$c_1(\mathcal{E}_+) = px + qy, \qquad p, q \in \mathbb{Z}. \tag{16}$$

Because $x^2 = y^2 = 0$,

$$\mathrm{ch}(\mathcal{E}_+) = 1 + px + qy + pq\, xy. \tag{17}$$

Choose the Callias chirality/orientation sign $\epsilon = \pm 1$ so that the boundary map is written

$$\mathrm{ch}(\mathrm{Ind}\, \mathscr{D}^+) = \epsilon \int_{S^2_\infty} \mathrm{ch}(\mathcal{E}_+). \tag{18}$$

This convention packages the possible overall sign from the spatial orientation and grading. Equations (17) and (18) give

$$\mathrm{rank}\, \mathrm{Ind}\, \mathscr{D}^+ = \epsilon p, \qquad c_1(\lambda_{\mathrm{C}}) = \epsilon pq\, y. \tag{19}$$

**Proposition 3.1** (Minimal rank-one Callias target)**.** *Within Eq.* (16)*, a single complex kernel with* $\lambda_{\mathrm{C}} \simeq \mathcal{O}(+2)$ *exists if and only if*

$$\epsilon p = 1, \qquad q = 2, \tag{20}$$

*assuming the cokernel vanishes and the kernel dimension stays constant.*

*Proof.* Rank one requires $\epsilon p = 1$. Substitution into Eq. (19) gives $c_1 = qy$, which equals $2y$ if and only if $q = 2$. Constant kernel dimension and zero cokernel identify the virtual determinant with the actual kernel line. □

The condition $p = 1, q = 0$ has the same pointwise index but a trivial determinant line. A doubled pair $q = +2$ and $q = -2$ has vanishing total determinant Chern class. These are decisive negative controls: counting a single zero mode at one moduli point does not compute a family Chern number.

### 3.3 Berry formula and finite template/control regression

When the kernel is an honest bundle with orthonormal frame $\Xi_A$, its determinant connection and Chern number are

$$\mathcal{A} = \mathrm{Tr} \int_{\Sigma_3} \Xi^\dagger \mathrm{d}_{\mathcal{M}} \Xi, \qquad \mathcal{F} = \mathrm{d}\mathcal{A} + \mathcal{A} \wedge \mathcal{A}, \tag{21}$$

$$c_1(\lambda_{\mathrm{C}}) = \frac{\mathrm{i}}{2\pi} \int_{\mathbb{CP}^1} \mathrm{Tr}\, \mathcal{F}. \tag{22}$$

For numerical regression define the spin-$k/2$ coherent vector

$$u_k(\theta, \phi)_r = \binom{k}{r}^{1/2} \cos^{k-r} \frac{\theta}{2} \sin^r \frac{\theta}{2} e^{-\mathrm{i}r\phi}, \qquad 0 \le r \le k. \tag{23}$$

It spans a smooth anti-Veronese Berry line of degree $+k$. With the standard degree-$k$ holomorphic structure this line is $\mathcal{O}(k)$; the embedding into the trivial finite Hilbert bundle itself is not claimed to be holomorphic. On an oriented triangulation, the gauge-invariant Bargmann flux is

$$\Phi_{abc} = \arg[\langle u_a, u_b \rangle \langle u_b, u_c \rangle \langle u_c, u_a \rangle], \qquad c_1^{\mathrm{disc}} = -\frac{1}{2\pi} \sum_{\triangle abc} \Phi_{abc}. \tag{24}$$

The minus sign is fixed by the anti-Hermitian convention $\mathcal{A} = u^\dagger \mathrm{d}u$ and $c_1 = (\mathrm{i}/2\pi) \int \mathcal{F}$; it is not chosen after seeing the result. The deterministic verifier uses 2208 triangles and obtains $c_1^{\mathrm{disc}} = 1.999999999999981$ for $k = 2$. It also tests $k = 0, 1, 4$.

The finite matrix template/control family

$$H_k(\theta, \phi) = \mathbf{1}_{k+1} - |u_k\rangle\langle u_k| \tag{25}$$

has an exact one-dimensional kernel, projector residual below machine precision, and nonzero spectrum equal to one. It verifies that a uniformly gapped family with $c_1 = 2$ is algebraically consistent. It is not a discretization of Eq. (12); the actual charged-soliton Yukawa data are still absent.

**Fail-closed gate 3.2** (Callias status)**.** The formula (19), its target class, and the Berry regression are complete. The same-model claims remain false because no operator $\mathscr{D}_m$, asymptotic eigenbundle $\mathcal{E}_+$, uniform mass gap, or regulator coupling of $\lambda_{\mathrm{C}}$ to the SQM wavefunction has been derived. In particular, a tangent kernel line does not automatically become an additional coefficient line in the quantum Dolbeault complex.

# 4 CPT, Serre duality, and the anti-canonical sector

## 4.1 Why the naive inverse line is insufficient

Quantize the conditional $B = +1$ SQM in a fixed canonical Dolbeault polarization with coefficient line

$$E_+ = \mathcal{O}(2), \qquad Q_+ = \sqrt{2}\,\bar{\partial}_{E_+}, \qquad \mathcal{H}_{0,+} = H^{0,*}(\mathbb{CP}^1, E_+). \tag{26}$$

Riemann–Roch gives

$$(h^0(E_+), h^1(E_+)) = (3, 0). \tag{27}$$

A raw sign reversal of a degree-two Berry/WZW line would suggest $E_{\mathrm{raw},-} = E_+^\vee = \mathcal{O}(-2)$, but

$$(h^0(\mathcal{O}(-2)), h^1(\mathcal{O}(-2))) = (0, 1). \tag{28}$$

Thus a signed integer $+2 \to -2$ alone does not construct the CPT triplet.

## 4.2 Fixed-polarization anti-canonical map

Serre duality on a complex curve states

$$H^0(\mathcal{M}, E)^* \simeq H^1(\mathcal{M}, K_\mathcal{M} \otimes E^\vee). \tag{29}$$

For $\mathcal{M} = \mathbb{CP}^1$, $K_\mathcal{M} = \mathcal{O}(-2)$. Therefore the coefficient line that realizes the conjugate triplet in the *same* Dolbeault polarization is

$$E_- = K_{\mathbb{CP}^1} \otimes E_+^\vee = \mathcal{O}(-2) \otimes \mathcal{O}(-2) = \mathcal{O}(-4), \tag{30}$$

and

$$(h^0(E_-), h^1(E_-)) = (0, 3), \qquad \mathrm{ind}\,\bar{\partial}_{E_-} = -3. \tag{31}$$

**Theorem 4.1** (CPT in a fixed Dolbeault polarization)**.** *Let the $B = +1$ zero modes be $H^0(\mathbb{CP}^1, E_+)$. An antiunitary CPT map that reverses chirality while retaining the same displayed complex polarization is implemented on zero modes by the Serre pairing if and only if the $B = -1$ coefficient line is $E_- = K_{\mathbb{CP}^1} \otimes E_+^\vee$. For $E_+ = \mathcal{O}(2)$, this is $\mathcal{O}(-4)$, not $\mathcal{O}(-2)$.*

*Proof.* The perfect Serre pairing

$$H^0(E_+) \times H^1(K \otimes E_+^\vee) \longrightarrow H^1(K) \simeq \mathbb{C} \tag{32}$$

gives an antilinear identification after choosing the Hermitian structure. Conversely, a perfect fixed-polarization pairing with the positive sector for all states identifies the negative cohomology with the Serre dual, fixing the coefficient line up to holomorphic isomorphism. Since $\mathrm{Pic}(\mathbb{CP}^1) = \mathbb{Z}$, its degree fixes it uniquely. □

An alternative description uses an anti-holomorphic involution and a conjugate polarization. When both sectors are rewritten in one fixed polarization, the canonical factor in Eq. (30) reappears. A physical charged two-colour CPT derivation must specify which polarization and fermion-measure regulator are used. The raw Callias or WZW line only supplies the $\mathcal{O}(-2)$ part; the additional canonical factor is independent UV data. The mathematics of Eq. (30) is closed, but the same-model physical map is not.

### 4.3 Finite SQM spectrum regression

For the Fubini–Study metric $\mathrm{d}s^2 = C\,\mathrm{d}w\mathrm{d}\bar{w}/(1+|w|^2)^2$, the $E_+ = \mathcal{O}(2)$ Dirac tower is

$$\lambda_{n,\pm} = \pm\frac{2}{\sqrt{C}}\sqrt{n(n+3)}, \qquad \mathrm{mult}(\lambda_{n,\pm}) = 2n+3, \qquad n \geq 1. \tag{33}$$

Serre duality gives the identical massive tower for the $E_- = \mathcal{O}(-4)$ block, with the three zero modes in opposite chirality [6]. The verifier builds four exact massive levels, checks

$$Q^2 = (Q^\dagger)^2 = 0, \qquad \{\Gamma, D\} = 0, \qquad H = \frac{1}{2}\{Q, Q^\dagger\}, \tag{34}$$

and confirms indices $+3$ and $-3$. This is a spectrum regression of the conditional SQM, not evidence for its microscopic realization.

## 5 Degree-five differential character and spatial transgression

### 5.1 The framed configuration-space line

Let the mixed-WZW target be

$$X = S^3 \times S^2, \qquad \widehat{\Omega}_n = n\,\widehat{u}_3 \smile \widehat{u}_2 \in \widehat{H}^5(X;\mathbb{Z}), \tag{35}$$

with normalized characteristic forms and classes

$$\int_{S^3}\omega_3 = 1, \qquad \int_{S^2}\omega_2 = 1, \qquad c(\widehat{\Omega}_n) = nu_3u_2. \tag{36}$$

Differential characters and their fiber integration define holonomy without choosing a five-dimensional extension [7, 8, 9].

Let $\widetilde{\mathcal{C}}_B$ be the framed configuration component before the gauge quotient and

$$\mathrm{ev} : \Sigma_3 \times \widetilde{\mathcal{C}}_B \longrightarrow X, \qquad (x, \Phi) \longmapsto \Phi(x), \tag{37}$$

be evaluation. Fiber integration gives the differential line class

$$\widehat{\mathcal{L}}_B := \int_{\Sigma_3} \mathrm{ev}^*\,\widehat{\Omega}_n \in \widehat{H}^{5-3}(\widetilde{\mathcal{C}}_B;\mathbb{Z}) = \widehat{H}^2(\widetilde{\mathcal{C}}_B;\mathbb{Z}). \tag{38}$$

Its curvature and characteristic class commute with fiber integration:

$$\frac{F_{\widehat{\mathcal{L}}_B}}{2\pi\mathrm{i}} = \int_{\Sigma_3} \mathrm{ev}^*(n\omega_3 \wedge \omega_2), \tag{39}$$

$$c_1(\widehat{\mathcal{L}}_B) = \int_{\Sigma_3} \mathrm{ev}^*(nu_3u_2). \tag{40}$$

This establishes the mathematically correct degree reduction. It does not yet give a line on the gauge quotient.

### 5.2 Gauge-basic descent: local and global conditions

Put $\mathcal{C}_B = \widetilde{\mathcal{C}}_B/\mathcal{G}$ and let $p : \widetilde{\mathcal{C}}_B \to \mathcal{C}_B$. The exact descent statement is

$$\widehat{\mathcal{L}}_B \in \mathrm{im}\left[p^* : \widehat{H}^2(\mathcal{C}_B;\mathbb{Z}) \longrightarrow \widehat{H}^2(\widetilde{\mathcal{C}}_B;\mathbb{Z})\right]. \tag{41}$$

For a free smooth action, this is equivalent to a $\mathcal{G}$-equivariant line with connection whose curvature is basic and whose vertical action is trivial. Infinitesimally,

$$\mathcal{L}_{\xi^\#}F = 0, \qquad \iota_{\xi^\#}F = 0, \qquad \xi \in \mathrm{Lie}\,\mathcal{G}. \tag{42}$$

These local equations are not sufficient for disconnected/large gauge transformations. One must also require trivial holonomy on every vertical loop and a trivial character on each stabilizer:

$$\mathrm{Hol}_{\widehat{\mathcal{L}}_B}(\gamma_g) = 1, \qquad \rho_\Phi : \mathrm{Stab}_{\mathcal{G}}(\Phi) \to U(1) \text{ is trivial.} \tag{43}$$

**Proposition 5.1** (Equivariant refinement criterion)**.** *A sufficient intrinsic construction of Eq.* (41) *is an equivariant differential refinement*

$$\widehat{\Omega}_n^{\mathcal{G}} \in \widehat{H}^5_{\mathcal{G}}(X;\mathbb{Z}) \tag{44}$$

*whose restriction is* $\widehat{\Omega}_n$*, followed by equivariant fiber integration. It produces a line on the quotient stack. Descent further to the coarse quotient requires the stabilizer condition in Eq.* (43)*.*

In physical terms, failure of Eq. (41) is the residual gauge anomaly of the WZW line. Cancellation of local triangle anomalies in a UV fermion table is necessary but does not alone compute the differential holonomy for every large gauge loop. The present model has not supplied Eq. (44) or Eq. (43); the descent flag remains false.

### 5.3 Pullback to the same soliton sphere

Suppose the same charged soliton actually has a collective embedding

$$\iota : \mathcal{M}_B \simeq \mathbb{CP}^1 \longrightarrow \mathcal{C}_B. \tag{45}$$

After descent, the line has degree

$$k_{\mathrm{eff}} := \int_{\mathbb{CP}^1} \iota^* c_1(\widehat{\mathcal{L}}_B^{\mathrm{quot}}) = \int_{\Sigma_3 \times \mathbb{CP}^1} \mathrm{ev}_\iota^*(n u_3 u_2). \tag{46}$$

**Theorem 5.2** (Necessary and sufficient WZW pullback gate)**.** *Assume gauge-basic descent and a fixed holomorphic structure on* $\mathcal{M}_B = \mathbb{CP}^1$*. The spatially transgressed line pulls back as* $\mathcal{O}(+2)$ *if and only if the integer in Eq.* (46) *is* $+2$*.*

*Proof.* Naturality of characteristic classes and fiber integration gives Eq. (46). Complex line bundles on $\mathbb{CP}^1$ are classified topologically and holomorphically by their degree because $H^2(\mathbb{CP}^1;\mathbb{Z}) = \mathbb{Z}$ and $\mathrm{Pic}(\mathbb{CP}^1) = \mathbb{Z}$ [10]. Hence degree $+2$ is equivalent to $\mathcal{O}(+2)$. ☐

For the factorized conditional family

$$\mathrm{ev}_\iota(x,m) = (g_B(x), s(m)), \qquad \int_{\Sigma_3} g_B^* u_3 = B, \qquad \int_{\mathbb{CP}^1} s^* u_2 = d, \tag{47}$$

Eq. (46) becomes

$$k_{\mathrm{eff}} = nBd. \tag{48}$$

Thus $(n, B, d) = (2, +1, +1)$ gives $+2$, whereas $d = 0$ gives zero and $n = 1$ gives only one. This is a useful sufficient ansatz and numerical regression. It is not a proof that the current scalar direction $s$ is the same normalizable collective coordinate as $m$.

For $B = -1$, the raw transgressed degree is $-2$. Section 4 shows that a fixed-polarization CPT triplet instead needs total coefficient degree $-4$. The missing canonical $-2$ must come from the CPT transformation of the vacuum line/fermion measure, not from changing Eq. (48).

# 6 Composition theorem and fail-closed gates

**Theorem 6.1** (Sufficient same-soliton AP-E3/AP-E4 composition criterion)**.** *Within the field content and fixed-polarization construction specified in this note, the AP-E3 level-two sector and AP-E4 SQM compose on one charged two-colour $B = 1$ soliton when all of the following conservative audit conditions are proved:*

*(C1) a stable $B = 1$ solution has a normalizable $\mathcal{M}_1 = \mathbb{CP}^1$ family, finite positive $L^2$ metric, and uniform noncollective bosonic gap;*

*(C2) the same four-dimensional action supplies either microscopic supercharges or independently derived emergent $N = 2$ Ward identities, together with a uniformly Fredholm Yukawa–Callias family;*

*(C3) its kernel bundle is $T^{1,0}\mathbb{CP}^1$, with the required Ward identity and no extra zero modes;*

*(C4) the actual families index has rank one and determinant Chern number $+2$, and the UV measure specifies whether that determinant line twists the wavefunctions;*

*(C5) the $B = -1$ regulator derives $E_- = K_{\mathbb{CP}^1} \otimes E_+^\vee = \mathcal{O}(-4)$;*

*(C6) the WZW character admits an equivariant differential refinement and satisfies every local and global gauge-basic condition;*

*(C7) the same-soliton integral $\int_{\Sigma_3 \times \mathbb{CP}^1} \mathrm{ev}^*(2u_3u_2) = +2$;*

*(C8) no gauge projection, anomaly constraint, or lower-energy mixing removes the conditional three-state sector.*

*Proof.* Conditions (C1)–(C3) supply the tangent-SQM closure. Condition (C4) is the Callias determinant criterion and prevents a pointwise-index substitution. Condition (C5) is the CPT theorem. Conditions (C6)–(C7) are the necessary and sufficient differential-cohomology descent and pullback criteria. Condition (C8) is required for the Hilbert-space sector to survive. Together they construct the claimed SQM. These are sufficient closure conditions for the present construction. They are not asserted to exhaust all possible emergent microscopic mechanisms. With the current evidence, each missing condition nevertheless invalidates the corresponding proposed derivation step, so this composition has not been established. □

| Claim | Status | Reason |
|---|---|---|
| Low-energy tangent-SQM characterization | closed | required low-energy data and a sufficient same-model audit are explicit |
| Bosonic underdetermination theorem | closed | constant-mass and winding-Yukawa completions share the bosonic sector |
| Same charged-soliton $\mathbb{CP}^1$ family | open | no solution family or orientational $L^2$ norm |
| Same-model $N = 2$ tangent fermion | open | no microscopic supercharges, emergent Ward-identity derivation, or Yukawa operator |
| Conditional Callias $c_1 = +2$ template | closed | boundary push-forward and Berry regression agree |

| Claim | Status | Reason |
|---|---|---|
| Actual same-model Callias index/determinant | open | asymptotic mass eigenbundle and uniform Fredholm gap absent |
| Mathematical anti-canonical CPT map | closed | Serre duality fixes $\mathcal{O}(-4)$ in a fixed polarization |
| Physical same-model CPT regulator | open | canonical vacuum-line factor not derived |
| Degree-five to degree-two fiber integration | closed | differential cohomology defines Eq. (38) |
| Gauge-basic descent | open | no equivariant refinement or large-gauge holonomy audit |
| Same-soliton pullback $\mathcal{O}(+2)$ | open | the exact integral is not computed on a physical moduli family |
| AP-E3/AP-E4 composition | open | conditions (C1)–(C8) are not all closed |
| Degree-one Route-E portal | not built | deliberately deferred until composition closes |
| Physics promotion | forbidden | conditional mathematics is not a microscopic derivation |

# 7 Deterministic verification and reproducibility

Run

```
PYTHONPYCACHEPREFIX=/tmp/python-cache \
python3 route_f/code/verify_ap_e4_same_soliton_callias_descent.py
```

The program emits

```
route_f/output/ap_e4_same_soliton_callias_descent.json
route_f/output/ap_e4_same_soliton_callias_descent.md
```

and hashes the TeX, bibliography, and verifier. Its deterministic tests are:

1. CP1 metric, tangent norm, and Levi–Civita chart covariance;
2. gauge-invariant Berry-Chern numbers $k = 0, 1, 2, 4$;
3. a uniformly gapped finite template/control kernel-family and its $c_1 = 2$;
4. Callias boundary push-forward, $q = 0$, and doubled-family controls;
5. finite $N = 2$ matrices, nilpotency, chirality pairing, and the $+3/-3$ Serre-dual indices;
6. the $\mathcal{O}(-2)$ one-state CPT negative control;
7. degree reduction $5 - 3 = 2$ and the $nBd$ transgression integer;
8. explicit fail-closed conditions for gauge descent and the same-model tangent theorem.

The checks encode consequences of the analytic derivations. They do not prove the existence of the missing microscopic structures.

## 8 Next calculations, in strict order

Several routes remain viable, but they should be pursued without weakening the gate:

1. **Supersymmetric same-model completion.** Specify a four-dimensional charged two-colour action with two preserved real supercharges, solve its $B = 1$ family, and derive the coupled boson–fermion quadratic operator. This is the direct route to (T1)–(T5).

2. **Non-supersymmetric Callias route.** Even without exact supersymmetry, specify the physical quark/Yukawa operator, calculate its asymptotic positive-mass eigenbundle on $S^2_\infty \times \mathcal{M}_1$, and test whether $(\mathrm{rank}, c_1) = (1, 2)$. A positive result gives a tangent Fermi bundle but still needs an emergent-SQM Ward-identity audit.

3. **Equivariant WZW route.** Construct $\widehat{\Omega}_2^{\mathcal{G}}$, compute vertical holonomies for representatives of small and large gauge transformations, then evaluate Eq. (46) on the explicit same-soliton family.

4. **CPT regulator route.** Compute the Bismut–Freed/Dai–Freed determinant-line transformation under $B \mapsto -B$, and show whether the canonical factor $K_{\mathbb{CP}^1}$ is supplied. A raw $-2$ result is not enough.

Only after one coherent mother model closes all composition conditions should a degree-one Route-E portal be written. Until then the correct endpoint is an explicit theorem of conditional composability plus a list of missing UV data, which this note now provides.

## 9 Machine-readable final flags

The verifier records, among others,

```
conditional_composition_sufficient_criterion_established=true,
low_energy_tangent_sqm_characterization_derived=true,
sufficient_same_model_audit_conditions_derived=true,
underdetermination_theorem_applies=true,
physical_tangent_fermion_same_model_derived=false,
callias_determinant_c1_same_model_computed=false,
conditional_callias_template_c1_plus_2_verified=true,
cpt_anti_canonical_map_mathematically_derived=true,
cpt_anti_canonical_map_same_model_derived=false,
wzw_spatial_fiber_integration_defined=true,
wzw_gauge_basic_descent_same_model_proven=false,
wzw_pullback_o2_same_model_proven=false,
ap_e3_ap_e4_composition_gate_closed=false,
degree_one_route_e_portal_built=false,
physics_promotion_allowed=false.
```

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